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Stone Piles as Kōura Habitat

Stone pile refuges in an Aotearoa New Zealand lake: early benefits for threatened crayfish but not its invasive predator

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Stone pile construction at Lake Rotoiti

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Abstract

Introduction: Kōura (*Paranephrops planifrons*), Aotearoa New Zealand’s endemic freshwater crayfish, have declined by up to 91% in Lake Rotoiti (Te Arawa Lakes, Rotorua), driven by predation from the invasive brown bullhead catfish (*Ameiurus nebulosus*) and thermal stratification that compresses suitable habitat into the shallow littoral zone, stressors that cannot be removed at the lake scale.

Objectives: We tested whether stone pile structures could provide kōura with structural refuge from catfish predation, and established a transferable construction methodology for lake managers.

Methods: Kōura and catfish were monitored at 20 locations (five stone piles, five control, and ten reference sites) over 18 months using a quasi before-after control-impact (quasi-BACI) design. Kōura occupancy was modelled within a Bayesian framework incorporating detection probabilities from baited traps and fyke nets, alongside a simulation-based power analysis.

Results: Stone pile construction produced no statistically significant change in kōura abundance, biomass, or occupancy relative to controls, but effects consistently trended positive across all models. Kōura use was strongly seasonal, falling in winter as kōura withdrew to deeper water at stone pile and control sites, while reference sites remained stable; catfish showed no comparable attraction. The study was underpowered, given high pre-construction kōura occupancy (81–89%) and only five site pairs.

Conclusions: Stone piles show early, consistent evidence of selectively benefiting kōura without benefiting its invasive predator, though the study lacked power to statistically confirm this trend.

Implications for Practice: Stone piles are most valuable when habitat compression and predation pressure peak together in summer and autumn, conditions expected to intensify under climate warming. This study establishes a tested, scalable construction and consenting methodology that lake managers can deploy as an interim refuge-based tool while catfish eradication remains unachievable at the lake scale. We recommend continued monitoring beyond three years, given the slow trajectories typical of crayfish colonisation, and trialling corridor-style structures linking refuge to deeper water rather than isolated piles, to improve winter resilience and support scaling toward population-level recovery. As catfish showed no comparable benefit from the structures in this trial, managers can pursue stone pile refuges for kōura without the risk of also benefiting this invasive predator.

1 Introduction

2 Freshwater biodiversity is declining faster than any other biome, with one quarter of freshwater
3 fauna now threatened with extinction (Sayer et al. 2025). Among the most endangered are
4 freshwater crayfish, with 32% of species assessed as threatened globally due to habitat loss,
5 invasive species, and declining water quality (Richman et al. 2015). As keystone species and
6 ecosystem engineers, crayfish play crucial roles in nutrient cycling, bioturbation, and food web
7 dynamics, and their loss can have cascading consequences for freshwater ecosystems (Collier
8 et al. 1997, Parkyn et al. 2001). In Aotearoa New Zealand, these pressures converge on
9 kōura (*Paranephrops planifrons* White, 1842), an endemic freshwater crayfish now facing rapid
10 population decline in lakes that have long supported some of the largest customary kōura
11 fisheries in the country (Hiroa 1921, Kusabs and Quinn 2009).

12 Kōura populations in Lake Rotoiti, located in the Rotorua Te Arawa Lakes region of the North
13 Island of Aotearoa New Zealand (hereafter Aotearoa), have declined by up to 91% over the past
14 two decades (Kusabs et al. 2026). This collapse has virtually ended customary harvest practices
15 for Te Arawa iwi (Māori tribal group) who are hunga tiaki (traditional custodians) of the lake
16 (Kusabs et al. 2015a, 2026). The decline in kōura numbers is driven by multiple interacting
17 stressors that compound over time in different parts of the lake (Lee et al. 2025, Kusabs et al.
18 2026). The invasive brown bullhead catfish (*Ameiurus nebulosus* (Lesueur, 1819); hereafter
19 catfish), confirmed in Lake Rotoiti in 2016 (Francis 2019, Grayling 2020), preys directly on
20 kōura and suppresses their foraging through non-consumptive ‘fear’ effects (Raven et al. 2026b).
21 Predation pressure from catfish is compounded by additional stressors that constrain where in
22 the lake kōura can seek refuge from this invasive predator (Kusabs et al. 2026).

23 During the day, kōura shelter in complex rocky habitat and in deeper water to avoid predation
24 (Jowett et al. 2008, Kusabs et al. 2015b), moving out of both at night to forage in the
25 productive littoral zone (Devcich 1979). The importance of deep-water refuge is shown in
26 Lake Taupō, where kōura persist at greater depth even in the presence of catfish because
27 the lake remains oxygenated below the thermocline (Kusabs and Taiaroa 2015, Verburg and
28 Albert 2019). In Lake Rotoiti, this deep refuge becomes anoxic during summer stratification,
29 compressing the zone of liveable habitat for kōura into the shallow littoral margins (Kusabs et
30 al. 2026), precisely where catfish are most active during summer (Dedual 2002, Blanchfield
31 et al. 2026). Dense beds of invasive macrophytes further restrict kōura movement through
32 this compressed zone and intensify oxygen depletion during die-off events (Vincent et al. 1984,
33 Hamilton et al. 2005, Kusabs and Quinn 2009). The dual pressures of catfish predation

1 and thermal stratification are expected to only strengthen with predicted climate scenarios
2 ([Woolway et al. 2021](#), [Lee et al. 2025](#)), making kōura habitat compression and predation
3 pressure progressively more severe.

4 The compression of suitable habitat and subsequent concentration of kōura into the littoral
5 zone makes the quality of nearshore lacustrine habitats increasingly important for remaining
6 populations. In stream environments, kōura are positively associated with structurally complex
7 habitats including bank undercuts, woody debris, leaf litter, and coarse substrates ([Parkyn
8 and Collier 2004](#), [Jowett et al. 2008](#), [Parkyn et al. 2009](#)). In lake environments, benthic
9 substrate characteristics are a stronger predictor of kōura distribution than nutrient enrichment
10 or predator abundance ([Kusabs et al. 2015b](#)), and kōura are positively associated with coarse
11 substrates and riparian vegetation ([Raven et al. 2026a](#)). Coarse rocky substrate provides
12 refugia from predators and shelter during moulting ([Streissl and Hödl 2002](#)). Experimental
13 studies confirm that shelter availability directly reduces catfish predation on crayfish ([Adams
14 2007](#), [Zhao et al. 2024](#)), and kōura increase refuge use in the presence of catfish compared
15 to native predators ([Raven et al. 2026b](#)), emphasising the importance of shelter availability
16 for kōura populations. Shelter availability is likely a limiting factor for kōura in Lake Rotoiti,
17 where the majority of the shoreline is sandy with dense beds of introduced macrophytes and
18 little rocky structure. Adding rocky substrate to soft-bottomed lake areas has been shown to
19 significantly increase crayfish abundance ([Johnsen and Taugbøl 2008](#)), and increasing structural
20 complexity in streams improves crayfish restocking success ([Huolila et al. 1997](#)). Deploying
21 habitat enhancement structures that add structural complexity in the littoral zone could
22 potentially be a practical way to restore kōura populations in Lake Rotoiti.

23 In this study habitat enhancement structures in the shape of stone piles were deployed along
24 the shoreline of Lake Rotoiti to investigate the ability of the structures in providing additional
25 habitat for kōura to live and hide from predators. Kōura presence, abundance, and biomass
26 were monitored at stone pile, control, and reference sites over 18 months. We predicted that
27 kōura occupancy, abundance and biomass would increase more rapidly at the stone pile sites
28 than at control sites, as intense catfish predation pressure in the littoral zone was expected
29 to compel kōura actively toward available refuge structures. We further predicted that kōura
30 abundance would continue to increase progressively after construction, at least within the first
31 18 months, and that catfish would instead show seasonal patterns of shoreline use rather than
32 a construction-driven trend ([Barnes 1996](#), [Dedual 2002](#), [Blanchfield et al. 2026](#)). Finally, we
33 predicted that stone pile sites would progressively converge toward kōura densities observed at
34 natural rocky reference sites, and that catfish would not exhibit a similar pattern.

Methods

Study location and construction

Lake Rotoiti (-38.03, 176.42) is a warm monomictic mesotrophic lake located in the Rotorua Te Arawa Lakes region of the North Island of Aotearoa (34 km², maximum depth 128 m). The lake stratifies for approximately nine months a year (?@fig-lake-conditions), with pronounced thermal stratification compressing oxygenated habitat into the shallow littoral zone during summer and autumn. Since 2008, nutrient-rich inflows from Lake Rotorua have been diverted toward the Kaituna River via the Ōhau Channel diversion wall to improve water quality in Lake Rotoiti (Hamilton et al. 2005, Prentice et al. 2026). Lake level is regulated by a weir at the outlet of the Kaituna River, providing a stable and controlled water level throughout the year. Catfish are present throughout the lake but occur in particularly high concentrations in Te Weta Bay on the Northwest side of the lake (Francis 2019). Five stone pile sites were selected alongside the lake shoreline based on accessibility and landowner approval (Figure 1). At each stone pile site, a paired control site was established 50 m along the shoreline, matched for habitat characteristics e.g., water depth, substrate type and vegetation cover. Five naturally occurring rocky shoreline locations, with each two monitoring sites 50 m apart were selected as reference. In total, 20 sites of each 150 m² (15 × 10 m) were monitored seasonally over 18 months. Physical characteristics of all sites are summarised in Table 1. Baseline monitoring was conducted in February 2025 prior to stone pile deployment, which occurred the following week. Stone piles were constructed from locally sourced quarry rock (Manawahe Stone and Quarry, ~25 km from the lake) with a diameter ranging from 85 - 315 mm (mean = 173 mm). At each of the five stone pile sites, three stone piles were arranged horizontally along the shoreline by hand from the shore. Stone pile design was informed by mesocosm experiments demonstrating kōura preference for enclosed stone structures (unpublished data). Stone piles were positioned approximately 4.3 m from shore at a mean depth of 0.49 m and with a mean area of 4.1 m².

Table 1: Physical characteristics of stone pile, control, and reference monitoring sites at baseline. Sites 1–10 are stone pile (odd) and paired control (even) sites; sites 11–20 are reference sites. Substrate index is a weighted composite of substrate particle sizes (higher values indicate coarser substrate more suitable for kōura). Slope is the gradient to the 20 m depth contour. Fetch is maximum effective fetch. Vegetation values are visually estimated percentage cover.

Site	Type	Habitat	Substrate index	Slope (m/m)	Fetch (km)	Riparian veg. (%)	Overhang (%)	Wood cover (%)
1	Stone pile	Rocky	5.20	0.12	4.76	0	10	0

2	Control	Rocky	4.40	0.11	4.79	70	0	0
3	Stone pile	Sandy	3.00	0.01	1.71	100	0	0
4	Control	Sandy	3.00	0.01	1.74	80	5	0
5	Stone pile	Sandy	2.40	0.01	0.61	10	0	0
6	Control	Sandy	3.00	0.01	0.55	0	0	0
7	Stone pile	Sandy	3.00	0.06	2.50	100	0	0
8	Control	Sandy	3.00	0.05	2.22	30	0	0
9	Stone pile	Sandy	3.35	0.08	3.59	0	0	3
10	Control	Rocky	5.00	0.09	3.74	20	0	0
11	Reference	Rocky	6.60	0.16	4.47	0	0	0
12	Reference	Rocky	6.20	0.12	4.17	100	100	0
13	Reference	Sandy	3.60	0.09	1.71	100	100	0
14	Reference	Sandy	3.50	0.11	1.65	100	100	0
15	Reference	Rocky	5.20	0.03	1.44	100	100	6
16	Reference	Rocky	5.20	0.04	1.40	100	100	2
17	Reference	Rocky	5.00	0.05	3.47	100	100	30
18	Reference	Rocky	6.20	0.05	3.46	100	100	7
19	Reference	Rocky	5.70	0.11	2.63	100	100	4
20	Reference	Rocky	5.70	0.13	2.87	100	100	0

¹ Source: [Article Notebook](#)

² Monitoring

³ Physical, chemical, and biological parameters were recorded at each site during each monitoring
⁴ round. Static site characteristics including substrate composition, slope to depth contour,
⁵ riparian vegetation, overhanging tree cover, and wood cover were visually estimated during
⁶ baseline monitoring only, as these were not expected to change substantially over the monitoring
⁷ period. Substrate composition was visually estimated by a single observer walking around
⁸ each site and using an underwater viewing chamber, categorising bedrock, boulders (>256
⁹ mm), cobble (64–256 mm), gravel (2–64 mm), sand (0.063–2 mm), mud (<0.063 mm), and
¹⁰ organic matter ([Wentworth 1922](#)). A substrate index, that indicates suitability for kōura, was
¹¹ calculated from these estimates ([Jowett and Richardson 1990](#), [Jowett et al. 2008](#)):

$$\begin{aligned}
 \text{Substrate index} = & 0.08 \times \text{bedrock} + 0.07 \times \text{boulders} \\
 & + 0.06 \times \text{cobble} + 0.04 \times \text{gravel} + 0.03 \times \text{sand} \\
 & + 0.02 \times \text{organic matter} + 0.01 \times \text{mud}
 \end{aligned}$$



Figure 1: Map of Lake Rotoiti in Bay of Plenty region of New Zealand. The monitoring sites are coloured by site type: stone pile sites red, control site yellow, and reference sites blue. Ten close-up maps show the placement of each of the sites.

1 Coarser substrates were assigned higher weights as they provide greater interstitial pore space
2 and stable structures used by kōura (Usio and Townsend 2000, Nyström et al. 2006).

3 After stone pile construction, measurements of the size and dimensions of each structure were
4 taken. The post-construction substrate index for stone pile sites was recalculated using each
5 site’s combined stone-pile footprint area, expressed as a percentage of the 150 m² site area
6 and treated as 80% cobble and 20% boulders displacing an equivalent area of sand, to reflect
7 their contribution to substrate complexity. Slope was assessed as the horizontal distance from
8 the shoreline to the 20 m depth contour, derived from bathymetric maps and measured in a
9 projected coordinate system (NZTM2000; EPSG:2193).

10 Seasonal variable parameters were recorded at every monitoring round and included aquatic
11 vegetation cover (emergent and submerged macrophytes). Chemical parameters included water
12 temperature (°C), dissolved oxygen (mg/L), and conductivity (S/cm) were measured using a
13 YSI ProSolo meter (Pro2030, YSI Inc., Yellow Springs, OH, USA). pH was measured using a
14 handheld pH meter (EC-PCTestr35, Eutech Instruments).

15 Kōura and fish were sampled every monitoring round using two single-winged, double-throated
16 fyke nets (600 mm high D-shaped entrance hoop) and two collapsible bait traps (480 × 250
17 mm; two 50 mm entrance openings) baited with catfish flesh (Joy et al. 2013, MacNeil et
18 al. 2025). Fyke nets were placed perpendicular to the shoreline outside of the stone pile
19 structures or at 10 m intervals, with bait traps positioned between them. All equipment was
20 deployed simultaneously at each site for a 24-h period. Captured kōura were measured for
21 orbital carapace length (OCL) using a digital calliper (0.01 mm precision; 150 mm) and for wet
22 mass using a digital scale (± 1 g; 5 kg capacity, Anko Brand, Kmart Group, Mulgrave, VIC,
23 Australia). Catfish and other fish species, were recorded for abundance, measured for total
24 length, and weighed for wet mass. Catch per unit effort (CPUE) and biomass per unit effort
25 (BPUE) were calculated for all kōura and fish based on the number of nets and traps deployed
26 at each site per monitoring round, standardised by soak time (Kusabs et al. 2015b). Kōura
27 weights were unavailable for 36 individuals because of scale failure and were therefore estimated
28 from a log –log length–weight regression fitted to individuals with both measurements (n
29 = 840), following standard approaches for weight length relationships in aquatic organisms
30 ?@fig-length-weight-ch5 (Froese 2006). Predicted values were back-transformed with a
31 lognormal bias correction factor $10^{0.5\sigma^2}$ (Quinn and Keough 2002).

$$\log_{10}(\text{Weight [g]}) = -2.67 + 2.684 \times \log_{10}(\text{OCL length [mm]})$$

1 Data analysis

2 Data analysis was a three-part process, the first assessed baseline equivalence between stone
3 pile, control, and reference sites before construction, the second assessed the impact of stone
4 pile construction and after-construction trajectory on kōura and catfish occupancy, abundance,
5 and biomass, and the third assessed whether stone pile sites converged toward reference sites
6 and if reference sites remained stable over time. All analyses were conducted in R (version
7 4.6.1; R Core Team (2026)) using the glmmTMB and emmeans packages (Brooks et al. 2017,
8 Lenth and Piaskowski 2017).

9 Covariate selection for all models followed a consistent procedure. Corrected Akaike Information
10 Criterion (AICc) values for each base model were compared against models with additional
11 covariates (temperature, season, substrate index, maximum effective fetch, or wave energy
12 proxy) added at a time, and a covariate was retained only when it reduced AICc by more than
13 two units relative to base model (Burnham and Anderson 2002). Robustness and sensitivity for
14 all models were assessed by evaluating uniform distribution, dispersion and outlier rate using
15 the DHARMA package (Hartig 2016). Across the full set of models, the Benjamini–Hochberg
16 correction method was applied to control the false discovery rate within each of the three
17 analysis components.

18 Site variation before stone pile construction was assessed using paired Wilcoxon signed-rank
19 tests to compare stone pile sites with their paired control sites, and unpaired Wilcoxon rank-sum
20 tests to compare stone pile sites with reference sites. This was done for response variables
21 (water quality, substrate, and kōura and catfish CPUE and BPUE) to confirm that the stone
22 pile and control sites were equivalent before construction, and to characterise the difference
23 between stone pile and reference sites. To further assess baseline equivalence, generalised
24 linear models (GLMs) were fitted to test whether kōura and catfish CPUE and BPUE differed
25 between site type (stone pile / control) and between the two reference sites (reference 1 /
26 reference 2) using a Tweedie family with a log link. Following the covariate selection procedure,
27 temperature was included in the catfish models comparing reference sites, as the base model
28 failed to converge and temperature was required for numerical stability.

29 The second component of the analysis tested construction impact relative to control sites using
30 a quasi-before-after, control-impact (quasi-BACI) design (Underwood 1994), and separately
31 tested after-construction trajectory of kōura and catfish populations. The design is quasi-
32 BACI because only a single pre-construction monitoring round was possible. The quasi-BACI
33 design compared one pre-construction monitoring round with six post-construction monitoring

1 rounds using generalised linear mixed effects models (GLMMs) to test the effect of stone pile
2 construction on kōura and catfish occupancy, abundance, and biomass. Time period (before
3 / after) and site type (stone pile / control) were included as an interaction term, monitoring
4 round as a covariate and site nested within control-impact pair as a random effect. Occupancy
5 was modelled using a binomial GLMM, and abundance and biomass using a Tweedie GLMM
6 with a log link; fetch was retained as a covariate in the catfish model. The post-construction
7 trajectory models used the same response variables, families, covariate, and random effect
8 structure, but with site type and monitoring round included as an interaction term in place of
9 the before / after comparison.

10 The third component of the analysis assessed whether stone pile sites converged toward reference
11 sites over time, and whether reference sites themselves remained stable. Kōura and catfish
12 abundance and biomass were tested using GLMMs with site type (stone pile / reference) and
13 monitoring round included as an interaction term, and site number as a random effect, using
14 a Tweedie family with log link; temperature was retained as a covariate in the catfish model.
15 Reference site stability was tested in GLMMs with site type (reference 1 / reference 2) and
16 monitoring round as an interaction term, and site nested within pair as a random effect, again
17 using Tweedie family with log link; season was retained as a covariate in the kōura model and
18 temperature was retained as a covariate in the catfish model.

19 To complement the GLMM abundance analysis, kōura site-level occupancy was estimated using
20 a two-method occupancy model fitted within a Bayesian framework. Because bait traps and
21 fyke nets provide independent and simultaneous detection opportunities at each site, true site
22 occupancy could be separated from imperfect and method-specific detection probabilities. A
23 custom Stan observation model was implemented using the brms package ([Bürkner 2017, 2018](#))
24 using cmdstanr ([Gabry et al. 2025](#)) to estimate true occupancy and detection probabilities
25 for baited traps and fyke nets via logit links. Standardised water temperature was included
26 as the covariate for detection probability. Three occupancy sub-models were fitted. A full
27 model across all monitoring rounds served as baseline comparison. A summer-only model and
28 a warm-season model (summer and autumn combined) were fitted to focus on the period of
29 highest thermal stratification, when kōura are known to disproportionately occupy nearshore
30 littoral habitats ([Kusabs et al. 2026](#)) and when catfish predation pressure is highest in the areas
31 where the stone piles are located ([Blanchfield et al. 2026](#)). The warm-season model maximises
32 ecological relevance, while retaining a tractable formula with season as an additive covariate.
33 All models were run with four chains of 4000 iterations (1500 warmup) and convergence was
34 assessed using $\hat{R}$ values (threshold < 1.01) and bulk effective sample sizes (threshold > 400)

1 (Vehtari et al. 2021). Divergent transitions were rare in all models (9 of 10,000 post-warmup
2 draws) and considered negligible given the otherwise strong convergence diagnostics.

3 A simulation-based power analysis was conducted to quantify the sample size required for
4 statistical detection of the BACI effect under the observed effect size and variance structure,
5 and to put the non-significant results in perspective relative to study design rather than true
6 absence of effect. For each analysis the occupancy and abundance, datasets were simulated
7 across a grid of scenarios varying in number of site pairs (3-30) and after-period monitoring
8 rounds (2-12). For the occupancy BACI, simulated binary detection data were generated
9 using values from the warm-season occupancy model posterior means, and each simulated
10 dataset was analysed using a logistic GLMM (lme4; Bates et al. (2015)). For the CPUE BACI,
11 simulated abundance data were generated using fixed effects and variance components from the
12 kōura abundance BACI model, using a log-normal approximation to the Tweedie distribution,
13 and each dataset was analysed using a linear mixed model fitted to a log-transformed CPUE.
14 Statistical power was estimated as the proportion of simulations (300 per scenario) in which
15 the model's 95% confidence interval for the BACI coefficient excluded zero in the positive
16 direction.

17 Results

18 Kōura presence was detected at stone pile sites within three months of construction, with
19 occupancy and abundance trending consistently higher at stone pile sites than paired control
20 sites throughout the 18-month post-construction period, though this difference did not reach
21 statistical significance. Results across all models and response variables are summarised in
22 Figure 2 and ?@tbl-models-summary-full-supp.

23 Baseline equivalence

24 Pre-construction, kōura abundance and biomass did not differ significantly between stone pile
25 and paired control sites (?@tbl-baseline-combined-supp), confirming baseline equivalences
26 ($p = 0.783$ and $p = 0.571$). Kōura abundance ($p = 0.968$) and biomass ($p = 0.968$) did not
27 differ between reference sites. Catfish abundance ($p = 0.004$) and biomass ($p = 0.002$) likewise
28 did not differ between stone pile and control sites pre-construction. However, catfish abundance
29 and biomass were significantly higher at Reference 1 than Reference 2 ($p = 0.004$ and $p =$

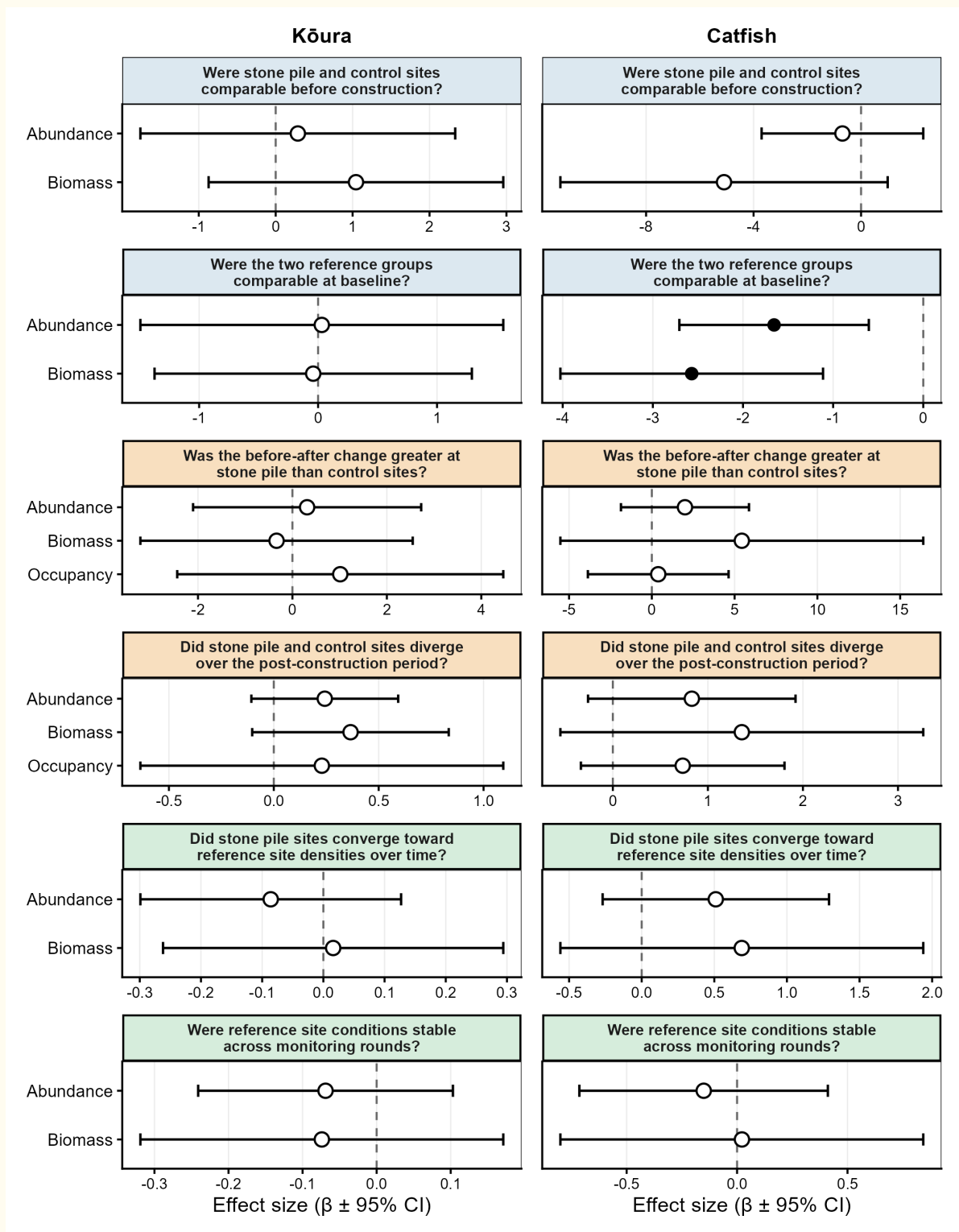


Figure 2: Effect sizes ($\beta \pm 95\% \text{ CI}$) for the key term of interest across all models, separated by species (columns) and comparison type (rows). Filled points indicate significant effects ($p < 0.05$); open points are non-significant. Each row has its own x-axis scale, effect magnitudes are not directly comparable across rows because each comparison tests a different contrast (step-change, per-round slope, or simple group difference), and occupancy is on the log-odds scale while abundance and biomass models use the Tweedie log link. Models with degenerate confidence intervals are excluded.

0.002, respectively), despite both sites having comparable rocky substrate. This suggests catfish distribution varies considerably between nearby sites with similar habitat characteristics.

Construction impact (BACI) and post-construction trajectory

Stone pile construction produced no detectable step-change in kōura abundance, biomass, or occupancy relative to paired control sites over the 18-month monitoring period following deployment (all $p_{\text{adj}} = 0.858$) (Figure 3). A matched-season comparison, comparing Summer 2025 (before construction) with Summer 2026 (one year after construction), produced the same conclusion (kōura abundance: $p = 0.582$). Post-construction trajectories did not diverge significantly between stone pile and control sites (all $p_{\text{adj}} = 0.215$). Kōura abundance declined significantly across post-construction rounds at control and stone pile sites ($\beta = -0.41$, $p = 0.004$), with the sharpest decline observed in Winter 2026 (Figure 4). Stone pile construction produced no detectable step-change or trajectory divergence in catfish abundance, biomass, or occupancy relative to control sites across any model or time period (Figure 2).

Kōura occupancy was high across all site types before construction, 81–89% in summer (?@tbl-psi-warm-supp). After construction, occupancy increased marginally at all sites, 89–94% in summer, thereby limiting our ability to detect a treatment effect. The BACI contrast to compare how much stone pile sites changed relative to control sites was slightly positive, but the credible intervals spanned zero with wide uncertainty (Figure 5), indicating no detectable additional increase at stone pile sites. However, this result likely reflects a measured ceiling rather than biological response. After-treatment contrast showed no significant difference between site types, but the direction of the effect was consistent with early habitat use: stone pile sites showed a positive contrast relative to control sites, while control sites showed a negative contrast relative to reference sites, and stone pile sites were near zero relative to reference sites (Figure 5).

The non-significant BACI results should be interpreted in the context of statistical power. For occupancy, power was near zero across all simulated scenarios (maximum 8%), reflecting the ceiling effect leaving no statistical room for a BACI signal (?@fig-power-curves-combined). Post-construction power analysis showed that achieving 80% power with the current five site pairs would require an effect size of approximately 1.8 log-units, almost 6 times the current observed effect (Figure 6). If the estimated positive trajectory of +0.24 log-units per seasonal round continues, the effect size may reach this detection threshold within approximately 6 additional monitoring rounds (~1.5 years).

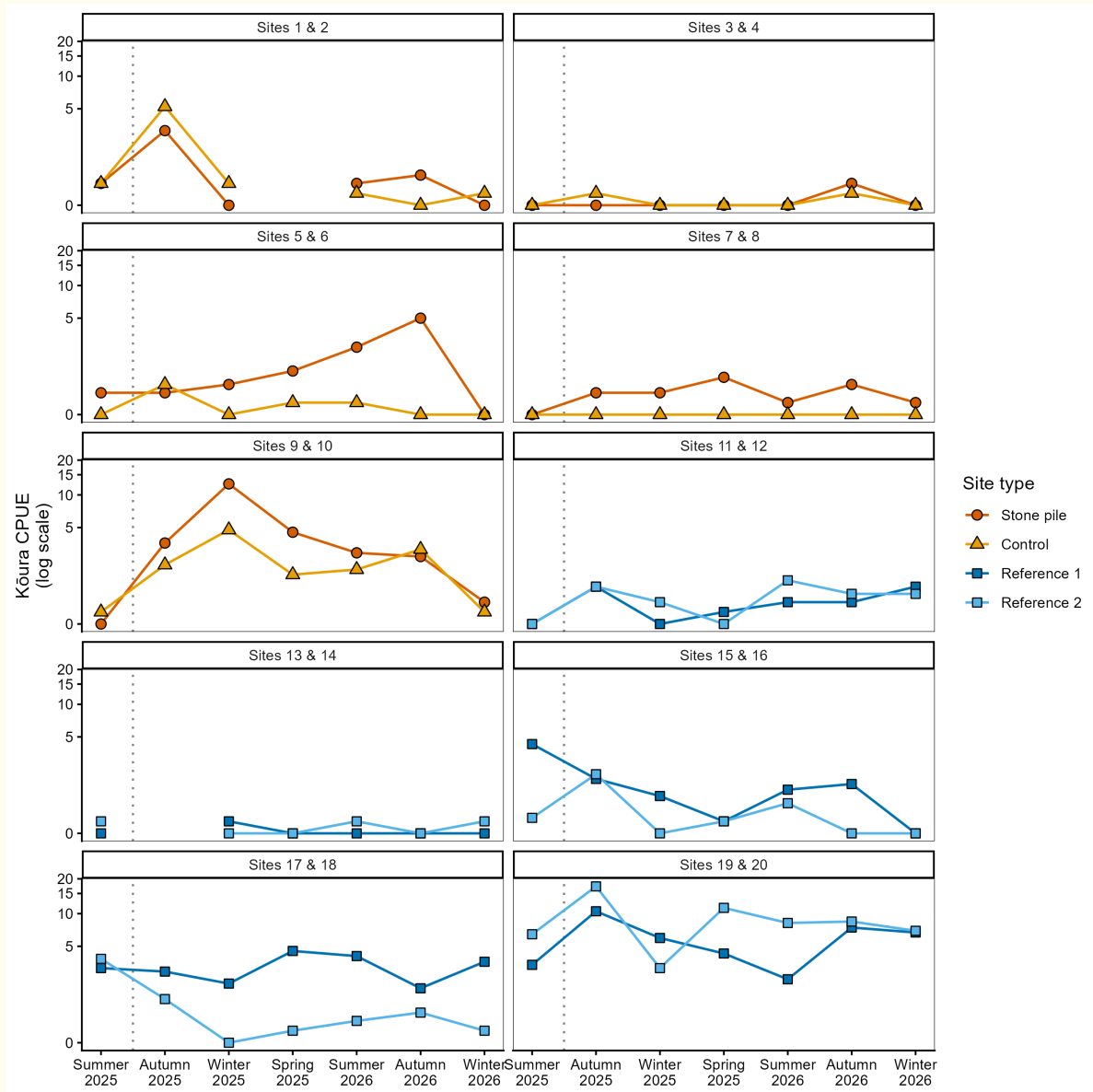


Figure 3: Kōura CPUE at individual site pairs across all monitoring rounds. Each panel shows one stone pile / control pair (sites 1–10) or reference pair (sites 11–20). Colour and shape indicate site type. Vertical dotted line marks stone pile construction. y-axis on $\log(1+x)$ scale.

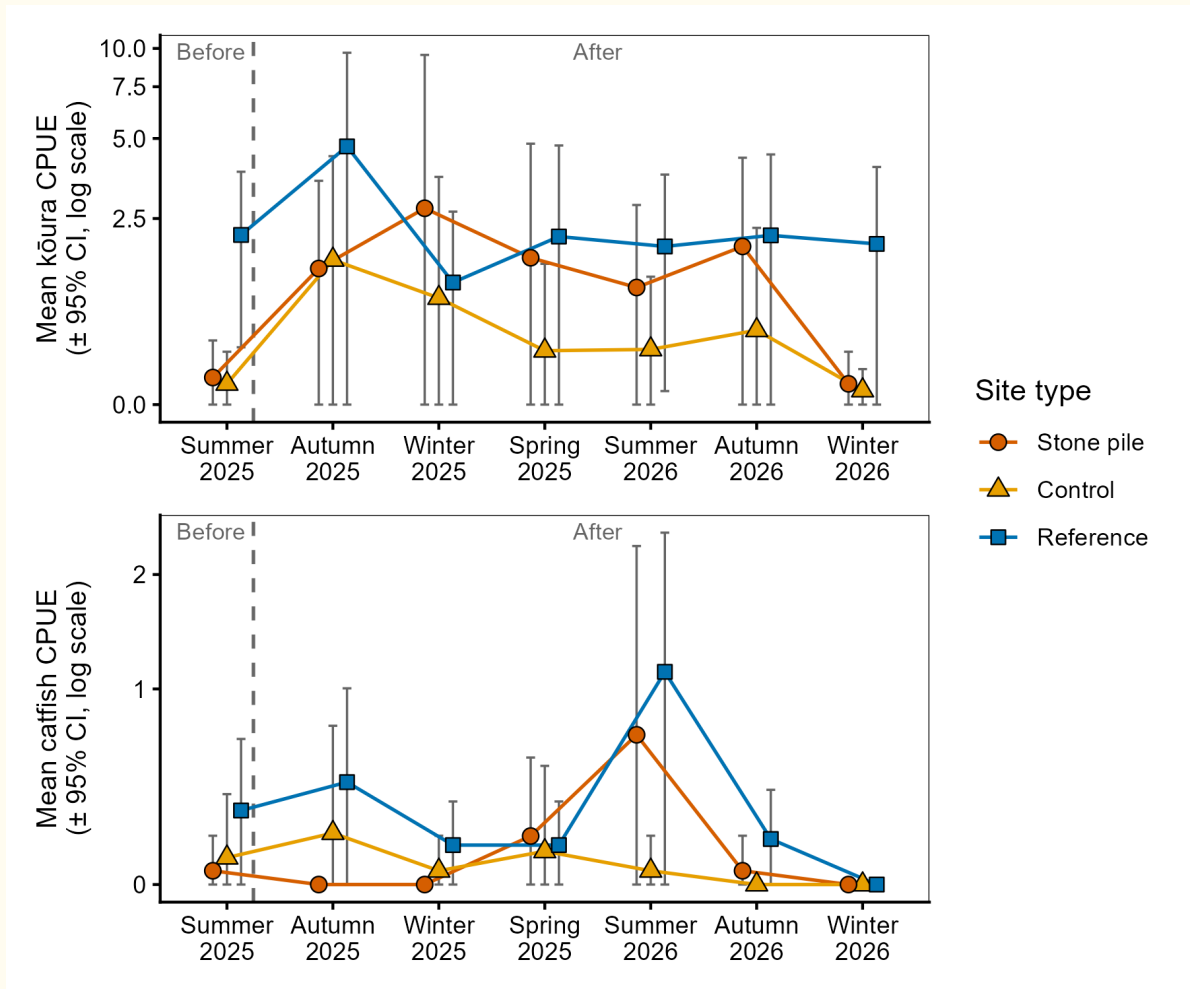


Figure 4: Mean CPUE per monitoring round ($\pm$ 95% CI, log scale) by site type for kōura (top) and catfish (bottom). Vertical dashed line distinguishes the pre-construction period (Before) from post-construction monitoring (After).

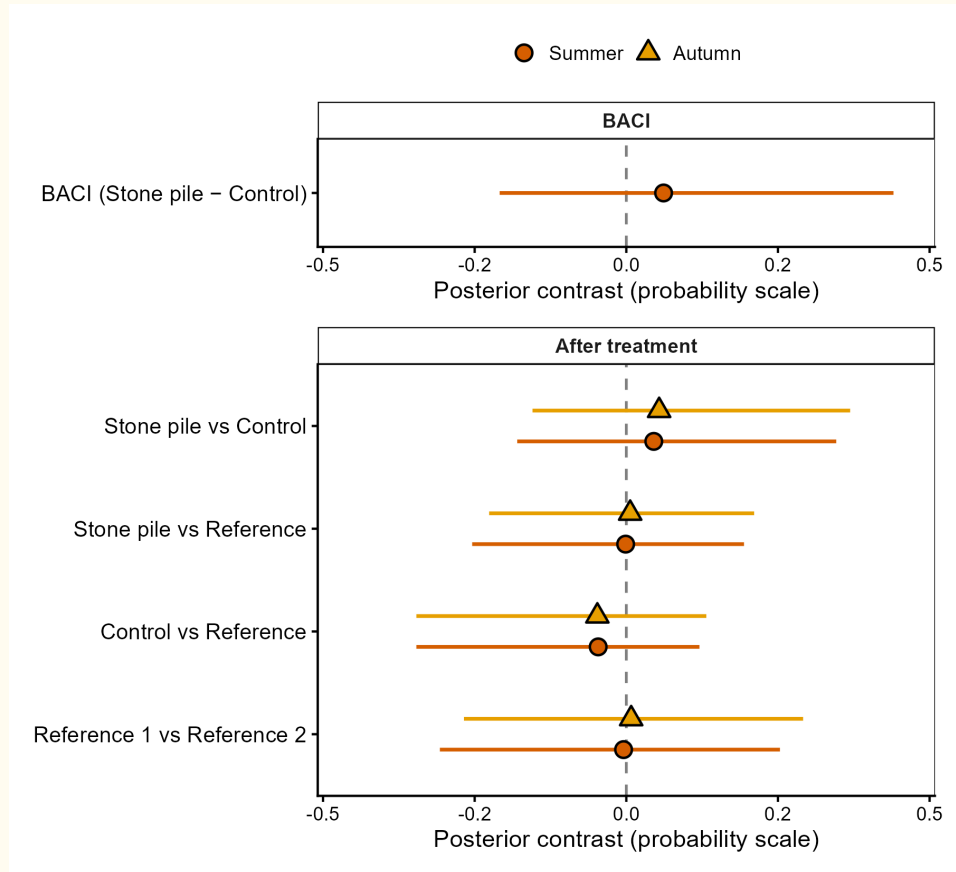


Figure 5: Posterior contrasts in kōura occupancy from the warm-season model. BACI contrast (stone pile minus control, before vs after construction). After-treatment contrasts between all site-type pairs. Points show posterior means with 95% credible intervals; contrasts overlapping zero indicate no detectable difference.

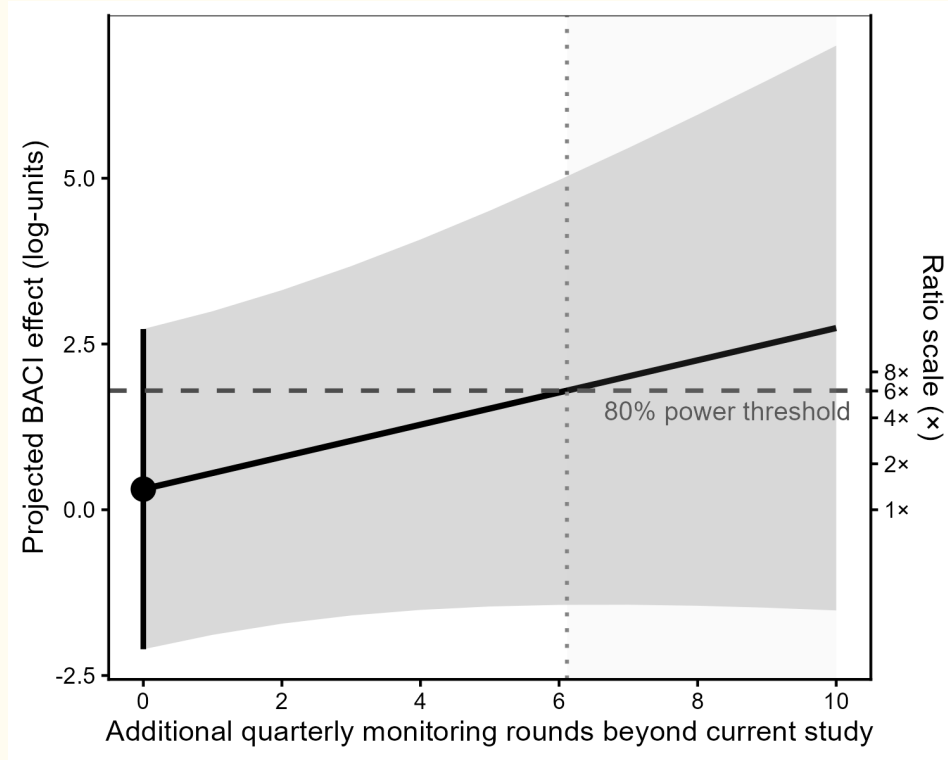


Figure 6: Projected kōura CPUE BACI effect size over additional monitoring rounds. The solid line shows the linear projection based on the estimated post-construction trajectory; shaded ribbon shows 95% uncertainty interval. Dashed horizontal line marks the detection threshold for 80% power with five site pairs.

1 Convergence toward reference sites and reference stability

2 Stone pile sites maintained a consistently lower kōura abundance than reference sites throughout
3 the post-construction period, with no significant change in this gap over time ($\beta = -0.09$, p
4 $= 0.429$; ?@fig-convergence-plot). Kōura abundance declined across monitoring rounds at
5 both stone pile and reference sites ($\beta = -0.09$, $p = 0.107$). The Winter 2026 monitoring round
6 showed a pronounced decline at stone pile sites (CPUE 0.7) while reference sites maintained
7 substantially higher abundance (CPUE 1.7; Figure 4), a within-round contrast not captured by
8 the linear trajectory model. Water temperature at Winter 2026 was approximately 1.5°C lower
9 than Winter 2025 but consistent across site types, ruling out temperature as a site-specific driver.
10 Visual inspection of stone pile surfaces during Winter 2026 monitoring revealed substantially
11 greater filamentous algal accumulation compared to earlier rounds (?@fig-stone-piles). Catfish
12 abundance and biomass showed no significant convergence or divergence between stone pile
13 and reference sites over time (Figure 2).

14 Discussion

15 Stone pile structures were deployed in the littoral zone of Lake Rotoiti to test whether physical
16 refuge could reduce predation pressure on kōura by invasive catfish. Stone piles produced no
17 statistically significant increase in kōura abundance, biomass, or occupancy relative to paired
18 control sites, but the direction of effect was consistently positive across every response variable
19 and model, and occupancy at stone pile sites converged with occupancy at natural rocky
20 reference sites by the first post-construction summer. Catfish showed no comparable response
21 to stone pile presence in any model, at any site type, or in any season, indicating that stone
22 piles selectively benefit kōura without providing equivalent refuge or foraging opportunity for
23 this invasive predator. A pronounced decline in kōura catch rates at stone pile and control sites
24 in Winter 2026, evidently driven by seasonal redistribution to deeper water rather than habitat
25 abandonment, showed that structural refuge value is itself seasonal and concentrated in the
26 warmer months when catfish predation pressure and habitat compression peak. Together, these
27 results and a formal power analysis indicate that the absence of statistical significance most
28 plausibly reflects limited study duration and sample size rather than genuine ineffectiveness of
29 the intervention.

1 Construction effect and statistical power

2 Kōura catch rates were highly variable across all site types throughout the monitoring period,
3 with wide confidence intervals, likely reflecting genuine biological variability in kōura distribution.
4 This variability may reflect seasonal activity patterns, detection probability changes across
5 monitoring rounds, and natural heterogeneity in littoral zone use, rather than stable site
6 associations. Despite this variability, the Bayesian occupancy model revealed a consistent
7 positive directional trend, with stone pile sites showing higher occupancy than control sites,
8 and occupancy comparable to natural rocky reference sites across both summer and autumn
9 monitoring rounds. Whether this reflects active refuge-seeking driven by catfish predation
10 pressure or gradual colonisation and population expansion into newly available habitats cannot
11 be determined from the current data, but the direction of the response was consistent with
12 the purpose of the structures, yet no statistically significant effect was detected within the
13 18-month monitoring period.

14 The absence of a significant effect is most likely explained by a combination of monitoring
15 duration, statistical power, and system dynamics rather than ineffectiveness of the structures.
16 Habitat establishment for freshwater crayfish can be inherently slow. Johnsen and Taugbøl
17 (2008) reported that crayfish abundance at a stone addition site took 13 years to reach reference
18 levels, and monitoring periods shorter than two years are often insufficient to detect significant
19 responses to habitat modification (Testa et al. 2011, Ruhí et al. 2012, Pilotto et al. 2018).
20 Following reconstruction of the Lake Rotorua lakefront, which is connected to Lake Rotoiti,
21 kōura showed no meaningful recolonisation after 28 months (Kusabs 2023). They attributed
22 slow recolonisation to persistent catfish predation pressure and distance from source population,
23 consistent with time scales observed here. The power analysis indicates that five site pairs
24 are insufficient to reliably detect effects of the magnitude observed here within 18 months,
25 establishing a minimum design threshold: future stone pile monitoring programmes should
26 plan for at least three years of monitoring and expect high pre-construction occupancy, leaving
27 little statistical room for a measurable treatment effect. The simultaneous elevation of kōura
28 CPUE across all site types in Autumn 2025 further complicated detection of an early treatment
29 effect in the BACI contrast.

30 Features of the system itself may also contribute to the weak response. Kōura abundance
31 is highest in lakes without catfish (Raven et al. 2026a), suggesting predation pressure may
32 suppress kōura populations well below the carrying capacity of existing habitat. If so, structural
33 habitat availability may not currently be the primary limiting factor, and a weak or delayed

1 response to habitat addition would be expected. Equally, if kōura are already at carrying
2 capacity of existing rocky habitat, stone piles add new capacity that will only be filled gradually
3 as the population expands, which is unlikely to be detectable within 18 months. In either
4 case, a delayed rather than absent population response would be expected. Isolated stone pile
5 structures may also be difficult for kōura to locate, since distance between isolated habitat
6 patches is a key determinant of colonisation success ([Hanski 1998](#)). Each of these explanations
7 point toward a delayed, rather than absent, population response, and continued monitoring
8 would be the appropriate next step rather than a reassessment of the intervention itself.

9 **Seasonal dynamics: the Winter 2026 decline**

10 Kōura CPUE at stone pile and control sites fell sharply in Winter 2026, returning to levels
11 close to the pre-deployment baseline from Summer 2025, while reference sites maintained
12 substantially higher abundance. This pattern is consistent with known seasonal migration
13 behaviour across lake depth profiles and kōura activity rates during cooler water temperatures
14 ([Devcich 1979](#)). When lake thermal stratification breaks down in winter, the hypolimnion
15 re-oxygenates and deep-water habitat becomes accessible, allowing kōura to redistribute along
16 the depth gradient ([Devcich 1979](#)). Reduced activity at low water temperatures independently
17 lowers the probability of trap capture ([Devcich 1979](#)), and although Winter 2026 was 1.5 °C
18 lower than Winter 2025, this difference was consistent across all site types and therefore cannot
19 explain why reference sites were substantially more resilient to the winter decline than the
20 stone pile sites.

21 Natural rocky reference sites in this study represent the ecological target for stone pile deploy-
22 ment. Reference sites had established kōura populations, connectivity to deeper waters, and
23 extensive refuge habitat, allowing kōura to maintain littoral presence through winter or return
24 rapidly from deeper refuges as conditions fluctuated. Stone pile sites are isolated structures with
25 less connectivity to deep and extensive refuge habitat, meaning kōura that withdrew in winter
26 had no quick pathway back to the structures. This difference likely explains why stone pile sites
27 maintained consistently lower kōura abundance than reference sites throughout the monitoring
28 period, with no significant convergence toward reference levels. However, kōura occupancy at
29 stone pile sites was comparable to that at reference sites after construction, suggesting that
30 kōura found and used the structures at similar frequency to natural rocky habitat but at lower
31 densities. This distinction between occupancy and abundance likely reflects the early stage of
32 colonisation and the small structural footprint of stone piles relative to the extensive natural

1 rocky habitat at reference sites. These differences are expected to narrow as biofilm succession
2 matures and kōura accumulate at the structures over time.

3 Extensive filamentous algae growth was observed on the stone pile structures in Winter 2026
4 but not to the same degree in Winter 2025, consistent with the seasonal pattern typical for
5 thermally stratified lakes, where winter mixing raises nutrient availability while cooler water
6 suppresses invertebrate grazing, allowing algal mats to accumulate on hard surfaces ([Wetzel](#)
7 [2001](#), [Bonacina et al. 2023](#)). The freshly quarried piles were free from biofilm at deployment and
8 by Winter 2026 supported a diverse meiofaunal assemblage, indicating active succession rather
9 than habitat degradation; algal cover would be expected to decline as grazer activity resumes
10 in spring and summer. Despite the near-zero trap catches in Winter 2026, night spotlighting at
11 two stone pile sites in August 2026 directly observed kōura sheltering within the structures,
12 confirming that low winter catch rates reflect reduced detectability rather than abandonment.
13 Together, these results indicate that stone pile refuge value is inherently seasonal — greatest
14 in summer and autumn, when thermal stratification compresses habitat into the littoral zone
15 and catfish predation peaks, and lowest in winter, when kōura redistribute to deeper water
16 regardless of structure type.

17 **Catfish: no evidence of attraction to stone piles**

18 Catfish showed their seasonal patterns in littoral zone use, but these were unrelated to stone pile
19 presence. Stone pile construction produced no evidence of catfish attraction or accumulation.
20 Even if catfish were to use stone piles to some degree, the ecological significance of this would
21 be limited given the extensive availability of existing suitable habitat for catfish in Lake Rotoiti,
22 particularly invasive macrophyte beds ([Barnes and Hicks 2003](#)). The single-site increase in
23 catfish CPUE during Summer 2026 was mirrored by an increase at reference sites and was
24 consistent with lake-wide seasonal redistribution into warm littoral waters rather than any stone-
25 pile-specific event ([Blanchfield et al. 2026](#)). Overall catfish catches remained low throughout
26 the study relative to targeted removal netting in Te Weta Bay, where highest catfish densities
27 are concentrated ([Francis 2019](#)). In autumn, catfish shift to mid-water movement across the
28 lake basin remaining above 15 m depth ([Blanchfield et al. 2026](#)), reducing predation pressure
29 on kōura in the littoral zone. This absence of attraction is an important management result
30 in its own right: habitat enhancement can be targeted at a threatened native species without
31 inadvertently improving conditions for its invasive predator.

1 Restoration design and future monitoring

2 Predator-prey interactions are shaped by habitat structure, which offers refuge to prey, influences
3 how easily predators locate and capture prey, and alters the costs and benefits of hunting for a
4 predator (Lennox et al. 2025). This framework applies directly to catfish-invaded lakes, where
5 habitat addition offers a practical lever on that interaction, and rocky environments have been
6 identified as natural multi-threat refugia, capable of simultaneously buffering predation, thermal
7 extremes, and invasive species pressure across a wide range of taxa and ecosystems (Selwood
8 and Zimmer 2020). In Lake Rotoiti, catfish and thermal stratification are the two primary
9 drivers of kōura decline, and neither can be addressed at the lake scale. Catfish eradication
10 is not currently achievable, and targeted catfish removal in high density areas reduces local
11 pressure without eliminating the lake-wide population (Dedual 2019). Thermal stratification,
12 meanwhile is expected to intensify under climate warming (Woolway et al. 2021, Lee et al.
13 2025). Where these stressors cannot be removed, habitat addition becomes one of the few
14 available management tools, and refugia become correspondingly more valuable (Selwood and
15 Zimmer 2020).

16 The stone piles deployed here were isolated structures, separated from each other and from
17 deeper water by soft sandy substrate. Isolated habitat patches support fewer species and are
18 more vulnerable to local extinction than connected habitats (Hanski 1998), and the Winter
19 2026 decline is consistent with this limitation: stone piles lacked the connectivity to deep refuge
20 that allowed reference sites to remain stable. Orienting future structures perpendicular to the
21 shoreline, extending through macrophyte beds and into deeper water, could create corridors
22 linking the littoral zone with deeper refugia, allowing kōura to move more freely between depth
23 ranges as conditions change, as they do at reference sites. This design has not been tested
24 but is a promising direction for future deployments, with the addition of potential benefit of
25 suppressing macrophyte growth along the corridor.

26 The dense algal fouling observed in Winter 2026 also raises a placement question. In sheltered
27 bays, stone piles are protected from wave damage, but algal growth and sediment accumulation
28 could progressively reduce crevice accessibility and refugia value; in more exposed locations,
29 natural scouring could limit fouling, but conditions may be too disturbed to provide effective
30 shelter (Parkyn and Collier 2004). This placement trade-off warrants direct investigation in
31 future deployments, and standardised photography of consistent stone faces each monitoring
32 visit would allow fouling development to be tracked alongside changes in kōura use.

1 Finally, the five stone pile structures deployed here cannot be expected to drive population-
2 level recovery across a lake of this size on their own. The spatial and temporal scale of the
3 ecosystem responses is typically greater than the scale of the interventions used to produce
4 them (Minns et al. 1996), and scaling from individual stone pile deployment to habitat addition
5 at population-relevant extents is a central challenge in restoration ecology (Ngwenya et al.
6 2025).

7 Beyond the ecological results, this study establishes a practical foundation for scaling habitat
8 enhancement in these lakes: the approvals, consents, and construction methodology needed
9 to deploy stone piles have now been tested, with evidence that doing so is unlikely to have a
10 counterproductive result by benefiting catfish. These outcomes give lake managers a practical,
11 tested option for future large-scale deployment; continued monitoring for at least three years is
12 recommended to determine whether the Winter 2026 decline was a one-off event and whether
13 kōura abundance at stone pile sites continues to build in subsequent spring and summer
14 seasons.

15 Stone pile deployment alone cannot reverse the trajectory of kōura decline in Lake Rotoiti,
16 but it directly addresses the current bottleneck: the shortage of structural refuge in a lake
17 where soft sediment dominates the shorelines and where the two primary stressors, catfish
18 predation and thermal habitat compression, cannot be controlled at the lake scale. Both
19 stressors are expected to intensify under continued climate warming (Woolway et al. 2021,
20 Lee et al. 2025), narrowing the band of habitat available to kōura and increasing, rather than
21 diminishing, the value of structural refugia over time. Establishing refugia before populations
22 decline further may be more effective than waiting until source populations fall below critical
23 densities for recolonisation (Morelli et al. 2016, Kusabs et al. 2026). The consistent positive
24 directional trend in occupancy, the rapid and clearly seasonal use of the structures, and the
25 developing biological community on the stone pile surfaces together suggest the foundations for
26 a population-level response by kōura are being laid, even though catfish showed no comparable
27 benefit from the structures. Whether this translates into a detectable population-level effect
28 will depend on continued long-term monitoring — the logical next phase of this restoration
29 programme.

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14 **Authors' contributions**

15 OVR, IAKK, RH, FJB, DÖ conceived and designed the research; OVR performed the fieldwork;
16 OVR analysed the data; IAKK obtained council and iwi consent and provided local ecological
17 knowledge; OVR drafted the manuscript; OVR, IAKK, RH, FJB, DÖ revised the manuscript.

18 **Use of Artificial Intelligence**

19 Use of Artificial Intelligence: Claude (Anthropic Sonnet 5.0) was used to assist with manuscript
20 editing and language refinement. All content was critically reviewed, and the authors take full
21 responsibility for the final manuscript.

22 **Declarations**

23 The corresponding author has declared that none of the authors has any competing interests.

1 Data availability statement

2 The code and derived data supporting the findings of this study are openly available on GitHub
3 (https://github.com/OlivierRaven/Stone_Piles_Rotoiti) and a rendered version of the analysis
4 is hosted at https://olivierraven.github.io/Stone_Piles_Rotoiti/. The primary raw dataset is
5 archived on Zenodo (<https://doi.org/...>). Peer review documents, including the cover letter
6 and reviewer responses, are available in the manuscript repository in the interest of open and
7 transparent science.

8 Supplementary information

9 *Supplementary figures are provided in the appendices.*

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